

Customer Feedback Theme Extractor (Multi-File Upload, Self-Serve Product)

Target duration: **14 days (2 weeks)**

Time commitment: ~1–2 hours per day

WEEK 1 — Core Functionality & Data Pipeline

Day 1: Project Setup & Architecture

Instructions

- Create a private GitHub repository
- Define a simple architecture:
 - Frontend: React or Next.js
 - Backend: FastAPI
 - AI layer: embeddings + LLM
- Set up a Python virtual environment
- Install core dependencies:
 - FastAPI
 - pandas
 - sentence-transformers or OpenAI embeddings
 - scikit-learn
 - pdfplumber / python-docx
- Create base folder structure:
- /frontend
- /backend
- /api
- /services
- /utils

Deliverables

- Repo initialized
 - Basic README with project overview
 - Backend app starts successfully
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Day 2: Multi-File Upload Support

Instructions

- Build frontend upload component:
 - Allow multiple files
 - Accept: CSV, XLSX, PDF, DOCX, TXT, JSON
- Add backend endpoint:
 - /upload
 - Validate file types
 - Save files temporarily
- Show upload progress in UI

Deliverables

- Users can upload multiple files
 - Backend receives and stores files
 - Error handling for unsupported formats
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Day 3: File Parsing & Normalization

Instructions

- Implement parsers for each file type:
 - CSV/XLSX → pandas
 - PDF → text extraction
 - DOCX → paragraph extraction

- JSON → recursive flattening
 - TXT/MD → direct read
- Convert all data into unified schema:
- feedback_id
- source_file
- raw_text

Deliverables

- All file types converted into a single list of feedback entries
- Console output showing total feedback count
- Unit test with mixed file types

Day 4: Text Cleaning & Quality Control

Instructions

- Remove:
 - Empty rows
 - Extremely short entries
- Normalize:
 - Whitespace
 - Encoding
- Add optional deduplication:
 - Embedding similarity threshold

Deliverables

- Clean feedback corpus
- Log showing removed duplicates
- Clean dataset ready for AI processing

Day 5: Embeddings & Clustering

Instructions

- Generate embeddings for all feedback entries
- Cluster using:
 - KMeans or hierarchical clustering
- Automatically determine number of clusters (or cap at 8–12)

Deliverables

- Cluster assignments per feedback item
- Saved embedding vectors
- Debug output showing cluster sizes

Day 6: Theme Label Generation

Instructions

- For each cluster:
 - Sample representative feedback
 - Send to LLM to generate:
 - Theme name
 - One-sentence description
- Store theme metadata

Deliverables

- Labeled themes
- Theme summary table:
 - Theme name
 - Mention count
 - Percentage of total feedback

Day 7: Sentiment Analysis

Instructions

- Run sentiment analysis on each feedback entry
- Aggregate sentiment by theme:
 - Positive
 - Neutral
 - Negative
- Store summary statistics

Deliverables

- Sentiment distribution per theme
- Overall sentiment breakdown
- Backend API returning sentiment data

WEEK 2 — UX, Reporting & Productization

Day 8: Quote Selection Engine

Instructions

- Select 3–5 representative quotes per theme
- Ensure:
 - Diversity of phrasing
 - Removal of PII patterns
- Store quotes with theme mapping

Deliverables

- Curated quote set per theme
- Quote preview endpoint

Day 9: Insights Dashboard (Frontend)

Instructions

- Build results page with:
 - Executive summary
 - Top themes ranked
 - Sentiment charts
 - Quotes by theme
- Keep layout simple and clean

Deliverables

- Fully functional insights dashboard
- Mobile-friendly layout
- Loading state and error state

Day 10: “What to Fix First” Recommendations

Instructions

- Prompt LLM using:
 - High-frequency themes
 - Negative sentiment weight
- Generate:
 - 3–5 actionable recommendations
- Display as bullet list

Deliverables

- Prioritized action list
- Clear, non-technical language

Day 11: PDF Report Generation

Instructions

- Build PDF export including:
 - Summary
 - Charts
 - Themes
 - Quotes
 - Recommendations
- Add simple branding header

Deliverables

- Downloadable PDF report
 - PDF matches dashboard insights
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Day 12: Demo Dataset & Example Run

Instructions

- Create a synthetic demo dataset:
 - Reviews
 - Survey responses
 - Support tickets
- Preload demo option in UI

Deliverables

- “Try with sample data” button
 - Marketing-ready screenshots
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Day 13: Hardening & Polish

Instructions

- Add:
 - File size limits
 - Timeout handling
 - Clear error messages
- Polish UI copy

Deliverables

- Stable demo
 - Smooth user experience
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Day 14: Deployment & Sales Readiness

Instructions

- Deploy backend (Render, Railway, Fly.io)
- Deploy frontend (Vercel, Netlify)
- Finalize README
- Write product description

Deliverables

- Live public URL
- Demo video or screenshots
- Product-ready build

Final Deliverables Checklist

- Multi-file upload support (6+ formats)
- AI-powered theme extraction
- Sentiment analysis

- Quote selection
- Interactive dashboard
- Downloadable PDF report
- Demo dataset
- Deployed web app

Resume-Ready Bullet Points (Copy/Paste)

- Built a production-ready, multi-format customer feedback analytics platform supporting CSV, Excel, PDF, DOCX, JSON, and text uploads
- Designed an end-to-end AI pipeline for feedback normalization, embedding-based clustering, theme extraction, and sentiment analysis
- Implemented automated executive-level reporting with interactive dashboards and downloadable PDF insights
- Delivered a self-serve analytics product translating unstructured customer feedback into actionable product recommendations